

EXPERIMENT-12

Implement Iris Flower Classification using K-Nearest Neighbors

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

print("=" * 70)
print("      IRIS FLOWER CLASSIFICATION USING KNN")
print("=" * 70)

uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
print("\nIRIS DATASET")
print("-" * 70)
print(data.head(10))
X = data.iloc[:, 1:5]
y = data.iloc[:, 5]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42
)
print("\nTraining Samples :", len(X_train))
print("Testing Samples :", len(X_test))
```

```

model = KNeighborsClassifier(n_neighbors=3)
print("\nTraining KNN Classifier...")
model.fit(X_train, y_train)
prediction = model.predict(X_test)
print("\nTraining Completed Successfully.")
print("\nActual Classes:")
print(list(y_test))
print("\nPredicted Classes:")
print(list(prediction))
print("\nAccuracy:")
print(round(accuracy_score(y_test, prediction) * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, prediction))

# New flower
sample = [[5.1, 3.5, 1.4, 0.2]]
result = model.predict(sample)
print("\n" + "-" * 70)
print("NEW FLOWER PREDICTION")
print("-" * 70)
print("Sepal Length :", sample[0][0])
print("Sepal Width :", sample[0][1])
print("Petal Length :", sample[0][2])
print("Petal Width :", sample[0][3])
print("\nPredicted Flower:")
print(encoder.inverse_transform(result)[0])

```

Sample Input:

```

... =====
                IRIS FLOWER CLASSIFICATION USING KNN
=====
Choose Files Iris.csv
Iris.csv(text/csv) - 5107 bytes, last modified: 8/5/2026 - 100% done
Saving Iris.csv to Iris.csv

```

Sample Output:

*** IRIS DATASET

```
-----  
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species  
0   1           5.1           3.5           1.4           0.2  Iris-setosa  
1   2           4.9           3.0           1.4           0.2  Iris-setosa  
2   3           4.7           3.2           1.3           0.2  Iris-setosa  
3   4           4.6           3.1           1.5           0.2  Iris-setosa  
4   5           5.0           3.6           1.4           0.2  Iris-setosa  
5   6           5.4           3.9           1.7           0.4  Iris-setosa  
6   7           4.6           3.4           1.4           0.3  Iris-setosa  
7   8           5.0           3.4           1.5           0.2  Iris-setosa  
8   9           4.4           2.9           1.4           0.2  Iris-setosa  
9  10           4.9           3.1           1.5           0.1  Iris-setosa
```

Training Samples : 105

Testing Samples : 45

Training KNN Classifier...

Training Completed Successfully.

Actual Classes:

[np.int64(1), np.int64(0), np.int64(2), np.int64(1), np.int64(1), np.int64(0), np.int64(1)]

Predicted Classes:

[np.int64(1), np.int64(0), np.int64(2), np.int64(1), np.int64(1), np.int64(0), np.int64(1)]

Accuracy:

100.0 %

Confusion Matrix:

```
[[19  0  0]  
 [ 0 13  0]  
 [ 0  0 13]]
```

NEW FLOWER PREDICTION

Sepal Length : 5.1

Sepal Width : 3.5

Petal Length : 1.4

Petal Width : 0.2

Predicted Flower:

Iris-setosa